Maximum Likelihood Estimation for Simple Linear Regression

COM3031 Probabilistic Machine Learning

1 Model Specification

We observe a dataset

$$\{(x_i, y_i)\}_{i=1}^{n},$$

where each output $y_i \in \mathbb{R}$ is generated according to a linear model with Gaussian noise:

$$y_i = wx_i + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

Equivalently, the conditional distribution of y_i given x_i is

$$y_i \mid x_i, w, \sigma^2 \sim \mathcal{N}(wx_i, \sigma^2).$$

2 Likelihood Function

The probability density of a single observation is

$$p(y_i \mid x_i, w, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - wx_i)^2 \right).$$

Assuming independence across observations, the likelihood of the full dataset is

$$p(\mathbf{y} \mid \mathbf{x}, w, \sigma^2) = \prod_{i=1}^{n} p(y_i \mid x_i, w, \sigma^2).$$

3 Log-Likelihood

Taking logarithms, we obtain the log-likelihood:

$$\ell(w, \sigma^2) = \log p(\mathbf{y} \mid \mathbf{x}, w, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - wx_i)^2.$$

4 MLE for the Weight w

We differentiate the log-likelihood with respect to w:

$$\frac{\partial \ell}{\partial w} = \frac{1}{\sigma^2}\sum_{i=1}^{n} x_i(y_i - wx_i).$$

Setting the derivative equal to zero:

$$\sum_{i=1}^{n} x_i(y_i - wx_i) = 0.$$

Expanding:

$$\sum_{i=1}^{n} x_i y_i - w \sum_{i=1}^{n} x_i^2 = 0.$$

Solving for w gives the MLE:

$$\boxed{\hat{w}_{\mathrm{MLE}} = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}}$$

5 MLE for the Noise Variance σ^2

Next, we differentiate the log-likelihood with respect to σ^2:

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^{n} (y_i - wx_i)^2.$$

Setting the derivative equal to zero:

$$-\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^{n} (y_i - wx_i)^2 = 0.$$

Solving for σ^2:

$$\boxed{\hat{\sigma}^2_{\mathrm{MLE}} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{w}_{\mathrm{MLE}} x_i)^2}$$

6 Interpretation

- The MLE for w minimises the sum of squared errors $\sum_{i=1}^{n} (y_i - wx_i)^2$, which explains the connection to least squares regression.
- The MLE for σ^2 is the mean squared error (MSE): $\hat{\sigma}^2_{\mathrm{MLE}} = \mathrm{MSE}$.